Lesson plan · for the teacher

A scheme of work

Programming · Level 4 · Lesson 4

Time
40 min

Materials
A single sheet for the term plan; The school's timetable and its list of devices; Darsi's course list for each level

Learning goals

- Plan progression: unplugged, blocks, then Python.
- Build a term of units with assessment points and a device timetable.
- Make it last: inclusion, a club, colleagues.

Activity: Plan a term together

1. Warm-up (5 min): With colleagues, list what pupils meet in computing now, grade by grade. Mark the gaps and the repeats.
2. Ladder (10 min): Draw a ladder on the board: unplugged and the robot, blocks, Python basics, then functions and algorithms. Place each grade on it.
3. Units (15 min): Pairs of colleagues plan one unit each: a big idea, an unplugged start, a plugged part, a project and an assessment point.
4. Timetable (5 min): Share the school's devices across classes week by week, with paper lessons in between.
5. Wrap-up (5 min): Agree who teaches which unit and where the shared folder lives. Then give the exit ticket.

This is a session for colleagues, not pupils: run it in a staff meeting or a planning day.

Discussion

- Where does our school repeat the same computing ideas?
- Which grade has nothing yet?
- Who will keep the scheme going if we leave?

Exit ticket

- Write one unit for our scheme: its big idea and its assessment point.

Try it at school tomorrow
With your colleagues, draft one term of computing on a single page: four to six units, the big idea of each, and how you will assess it. Ask one colleague to teach a unit with you.

Practice questions · for you

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1 **You lead computing, and you are moving to another school next year. What best keeps computing going?**

- ◯ A shared scheme, trained colleagues and a club
- ◯ Keep the plans on your own phone
- ◯ Teach extra lessons before you leave

2 **Put the stages of a computing progression in order, from the youngest pupils up.**

- ☐ Blocks and very short programs
- ☐ Unplugged games and the robot's cards
- ☐ Python: functions, lists and algorithms
- ☐ Python: print, variables, if and loops

3 **Which stage of your scheme is this task for?**

```
for i in range(1, 11):
    print(i * 7)
```

- ◯ University
- ◯ Lower secondary (12 to 15), after loops
- ◯ Primary (9 to 12), the first term

4 **Every unit in a scheme of work should have an assessment point.**

- ◯ True
- ◯ False

5 **Match each unit to its big idea.**

The robot	• • • •	Many values in order, and loops over them
Decisions	• • • •	Exact instructions, in order, and loops
Lists	• • • •	Algorithms judged by the work they do
Searching and sorting	• • • •	if and else, tested at the boundaries

✎ **Which grade in your school has no computing at all today? Write the first unit you would give them, its big idea and its assessment point.**

Check your answers

A scheme of work

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1 **A shared scheme, trained colleagues and a club**

Shared plans and trained colleagues mean computing doesn't depend on one person.

2 **Top to bottom: 2, 1, 4, 3**

Each stage needs the one before: exact instructions and loops first, unplugged; then the same ideas in blocks; then in Python; then the bigger structures built from them.

3 **Lower secondary (12 to 15), after loops**

It needs for and range, so it comes after pupils have met loops unplugged and with the robot, in their first year of Python.

4 **True**

Without a planned check, you only find out at the end of the year who didn't learn.

5 **The robot — Exact instructions, in order, and loops**
Decisions — if and else, tested at the boundaries
Lists — Many values in order, and loops over them
Searching and sorting — Algorithms judged by the work they do

One big idea per unit keeps it focused, and tells you what to assess at its end.